WEATHER PREDICTION AI

A PROJECT REPORT

Submitted by

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ODISHA

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CERTIFICATE

DEPARTEMENT OF COMPUTER SCIENCE AND ENGINEERING

SCHOOL OF ENGINEERING AND TECHNOLOGY

BHUBANESWAR CAMPUS

BONAFIDE CERTIFICATE

Certified that this project report **Weather Prediction AI** work of our Group with all who carried out the project work under our supervision. This is to further certify to the best of my knowledge, that this project has not been carried out earlier in this institute and the university.

SIGNATURE

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Department of Electronics and Communication Engineering

Certified that the above mentioned project has been duly carried out as per the norms of the college and statutes of the university.

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DECLARATION

We hereby declare that the project entitled **"WEATHER PREDICTION AI"** submitted for the **"Machine Learning Using Python"** 6th semester B- tech in Computer Science and Engineering is our original work and the project has not formed the basis for the award of any Degree / B- tech or any other similar titles in any other University / Institute.

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ABSTRACT

Weather forecasting plays a crucial role in modern society by helping individuals, industries, and governments make informed decisions. This project focuses on developing a weather forecasting system using Python and machine learning techniques. Traditional forecasting methods rely heavily on numerical weather prediction models, which require high computational power and complex simulations. In contrast, this project explores a data-driven approach, where historical weather data such as temperature, humidity, wind speed, and precipitation are analyzed to predict future weather conditions. By leveraging Python libraries and machine learning algorithms, the system aims to provide accurate, efficient, and cost-effective forecasting solutions.

The project involves collecting and preprocessing weather datasets from reliable sources. Data cleaning, normalization, and feature selection are performed to ensure the quality of input data. Various machine learning models such as Linear Regression, Decision Trees, Random Forest, and Support Vector Machines are implemented and compared. These models are trained on historical data to identify patterns and relationships between different weather parameters. The use of Python libraries like NumPy, Pandas, Scikit-learn, and Matplotlib simplifies data handling, model training, and visualization, making the system both powerful and easy to implement.

A key objective of this project is to evaluate the performance of different machine learning models in predicting weather conditions. Metrics such as accuracy, mean squared error, and R-squared values are used to measure model effectiveness. The system is designed to be scalable and adaptable, allowing the integration of additional data sources or advanced algorithms like neural networks in the future. Visualization tools are also incorporated to present predictions in an understandable format, helping users interpret weather trends easily.

In conclusion, this project demonstrates how machine learning and Python can be effectively used to build a practical weather forecasting system. It provides an accessible alternative to complex traditional models while maintaining reasonable accuracy. The project also serves as an educational platform for understanding the application of machine learning in environmental science. Future enhancements may include real-time data integration, deep learning models, and improved prediction accuracy, making the system more robust and reliable for real-world applications.

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CHAPTER 1
INTRODUCTION

1.1 Evolution Of Weather Forecasting Platform.

The evolution of weather forecasting platforms has transformed from basic observation methods to advanced scientific systems. In ancient times, people relied on natural signs like cloud patterns, wind direction, and animal behavior to predict weather, which was often unreliable. During the 19th and early 20th centuries, the use of scientific instruments such as thermometers, barometers, and anemometers improved data collection. The invention of the telegraph allowed faster communication of weather data, while the introduction of satellites and radar systems enabled large-scale monitoring of atmospheric conditions, significantly enhancing forecasting accuracy.

With the rise of computers, numerical weather prediction models were developed, using mathematical calculations to simulate weather patterns. Over time, advancements in computing power made these predictions faster and more precise. In the modern era, weather forecasting platforms use artificial intelligence, machine learning, and big data to analyze real-time information from satellites and sensors. These platforms are now easily accessible through mobile apps and websites, providing instant updates, alerts, and interactive maps. As a result, weather forecasting has become more reliable and efficient, helping people and governments prepare for daily activities and natural disasters.

1.2 Importance Of Weather Forecasting Platform.

Weather forecasting platforms play a vital role in modern society by helping people make informed decisions in their daily lives. Accurate weather predictions allow individuals to plan activities such as travel, farming, outdoor events, and work schedules. For farmers, these platforms are especially important as they provide information about rainfall, temperature, and seasonal changes, helping in crop planning and irrigation management. In addition, weather forecasts support transportation sectors like aviation, shipping, and road travel by reducing risks caused by extreme weather conditions such as storms, fog, or heavy rainfall.

Moreover, weather forecasting platforms are crucial for disaster management and public safety. They provide early warnings about natural disasters like cyclones, floods, heatwaves, and thunderstorms, allowing governments and emergency services to take preventive measures and save lives. These platforms also help industries like energy, construction, and tourism to operate efficiently by adapting to weather changes. With the advancement of technology, modern forecasting platforms offer real-time updates through mobile apps and websites, making information easily accessible to everyone. Overall, weather forecasting platforms enhance safety, improve economic planning, and contribute to better preparedness against environmental challenges.

1.3 Overview On Our Glass Weather AI.

Glass Weather AI is an advanced weather forecasting platform developed by our team to provide accurate, real-time weather information in a user-friendly and intelligent way. The system displays a wide range of weather parameters such as temperature, humidity, wind speed, atmospheric pressure, rainfall, AQI and air quality. It collects data from reliable weather APIs and sensors, processes it using modern technologies, and presents it through an interactive and visually appealing interface. The platform is designed to be accessible on web and mobile devices, allowing users to easily check weather updates anytime and anywhere. Its clean dashboard and dynamic charts help users quickly understand current conditions as well as future forecasts.

A key feature of Glass Weather AI is its integrated AI chatbot, which enhances user interaction and experience. The chatbot can answer weather-related queries, provide personalized forecasts, explain weather conditions in simple language, and even give suggestions based on the weather (such as travel or clothing advice). By using artificial intelligence and machine learning, the system can improve prediction accuracy and deliver smarter insights over time. Overall, Glass Weather AI is a modern, intelligent, and efficient solution that combines real-time data, advanced analytics, and conversational AI to make weather forecasting more accessible, informative, and user-centric.

1.4 Project Motivation.

The motivation behind developing Glass Weather AI is to create a smart, user-friendly, and reliable weather forecasting system that overcomes the limitations of traditional platforms. Many existing systems provide complex data that is difficult for users to understand or lack personalized insights. Our goal is to simplify weather information using an intuitive interface and an AI-powered chatbot that can interact with users and provide clear, customized responses. By integrating real-time data, advanced analytics, and conversational AI, Glass Weather AI aims to improve decision-making, enhance safety, and make weather forecasting more accessible and meaningful for everyone.

1.5 Real World Applications Related To weather Forecasting.

Weather forecasting has numerous real-world applications across different sectors. In agriculture, farmers rely on forecasts to plan sowing, irrigation, and harvesting, which helps improve crop yield and reduce losses. In transportation, including aviation, shipping, and road travel, weather updates ensure safety by helping avoid storms, fog, and extreme conditions. Disaster management agencies use forecasting to predict natural calamities like cyclones, floods, and heatwaves, enabling early warnings and saving lives and property.

In addition, weather forecasting plays a key role in industries such as energy, construction, and tourism. Power companies use forecasts to manage electricity demand, especially during extreme temperatures. Construction projects depend on weather conditions for scheduling work safely. The tourism sector uses forecasts to plan travel activities and ensure a better experience for tourists. Moreover, everyday users benefit from weather apps to plan daily routines, clothing, and outdoor activities.

CHAPTER 2
FORMULATION OF THE PROJECT

2.1 Numeric Weather Predictions & Foundations.

Numerical Weather Prediction (NWP) systems are the foundation of modern weather forecasting. Instead of relying only on observation or experience, NWP uses the laws of physics to predict how the atmosphere will change over time. These systems solve complex mathematical equations that describe atmospheric processes such as temperature changes, air pressure, wind movement, humidity, and energy transfer. The atmosphere is divided into a three-dimensional grid, and calculations are performed at each grid point to simulate how weather conditions evolve. Because these equations are highly complex and involve millions of calculations, powerful computers are required to process them efficiently and produce forecasts within a useful time frame.

Organizations like the National Oceanic and Atmospheric Administration (NOAA) highlight that accurate numerical prediction depends heavily on two key factors: high-quality observational data and strong computational resources. Observations are collected from satellites, weather stations, radars, ocean buoys, and aircraft. This data is then processed through a technique called data assimilation, which ensures that the model starts with the most accurate representation of the current atmosphere. If the input data is incomplete or inaccurate, the forecast quality will decrease significantly. Additionally, high-performance computing systems are essential because they allow models to run at higher resolutions and include more detailed physical processes, leading to better predictions.

Similarly, the European Centre for Medium-Range Weather Forecasts (ECMWF) operates one of the most advanced global forecasting systems in the world. ECMWF explains that its success comes from the integration of three major components: advanced atmospheric physics, sophisticated data assimilation techniques, and state-of-the-art supercomputing. The physical models represent real-world processes such as cloud formation, radiation, precipitation, and turbulence. Data assimilation continuously updates the model using new observations, improving accuracy over time. Supercomputers enable these complex models to run quickly and efficiently, producing reliable forecasts for several days or even weeks ahead.

Overall, these references clearly show that physics-based modeling is the backbone of operational weather forecasting. NWP systems combine scientific principles, real-time data, and computational power to simulate the atmosphere and predict future conditions. This approach has significantly improved forecast accuracy and plays a crucial role in weather-related decision-making, disaster preparedness, and climate research worldwide.

2.2 Data Assimilation & Forecast Limitations.

In the context of Indian weather forecasting, data assimilation plays a crucial role in improving the accuracy of Numerical Weather Prediction (NWP) models. India has a diverse climate system influenced by monsoons, cyclones, and regional variations, making forecasting highly complex. Observations are collected from ground-based weather stations, satellites like INSAT, Doppler weather radars, ocean buoys, and aircraft. These observations are integrated using data assimilation techniques to create an accurate initial state of the atmosphere before

running forecast models. Organizations such as the India Meteorological Department (IMD) use this approach to improve predictions of monsoon rainfall, cyclone tracks, and extreme weather events. Accurate initialization is especially important in India, where small changes in atmospheric conditions can significantly affect weather outcomes.

However, despite advancements, weather forecasting in India still faces several challenges. One major issue is the uneven distribution of observation systems, particularly in remote and rural areas such as mountainous regions (Himalayas) and vast oceanic zones. This leads to gaps in data, which can reduce forecast accuracy. Additionally, NWP models rely on approximations of physical processes like cloud formation and rainfall, which may not fully capture the complexity of India's tropical climate. Another limitation is the chaotic nature of the atmosphere, where small errors in initial conditions can lead to large differences in forecasts over time. These factors make it difficult to provide highly precise and localized weather predictions across all regions of India.

To address these limitations, Indian forecasting systems are increasingly incorporating statistical post-processing and machine learning techniques. Machine learning models analyze historical weather data and model outputs to identify patterns and correct systematic errors (biases) in forecasts. They are also used for downscaling, which means converting large-scale model outputs into more detailed, location-specific predictions. For example, ML can help provide accurate rainfall forecasts at district or village levels, which is highly beneficial for agriculture and disaster management. Rather than replacing NWP, machine learning acts as a complementary tool, enhancing forecast accuracy and providing more user-friendly and localized insights, which is essential for a country like India with diverse weather conditions**.**

2.3 Machine Learning for Weather Forecasting.

Recent survey literature shows that machine learning (ML) and deep learning are playing an increasingly important role in weather and climate forecasting. Traditionally, forecasting relied mainly on physics-based Numerical Weather Prediction (NWP) models, but newer research highlights how data-driven approaches are becoming powerful additions. A 2025 survey published in *Frontiers of Computer Science* explains that ML techniques can learn patterns directly from large volumes of historical and real-time weather data. These models are especially useful for identifying complex, non-linear relationships in atmospheric processes that are difficult to represent using traditional equations. The survey also emphasizes the growing trend of combining ML with physical laws, ensuring that predictions remain scientifically consistent while benefiting from data-driven accuracy.

Similarly, a 2022 review in *Applied Sciences* provides a systematic analysis of various ML-based weather forecasting methods. It examines techniques such as regression models, decision trees, random forests, and deep learning architectures like neural networks and recurrent neural networks (RNNs). The review compares their performance across different forecasting tasks, including temperature prediction, rainfall estimation, wind speed analysis, and extreme weather detection. It highlights that ML models often perform very well in short-term

forecasting and localized predictions because they can quickly adapt to specific regional patterns. However, it also notes that their performance depends heavily on the quality and quantity of available data.

2.4 Interpretability and Practical Constraints.

In real-world weather forecasting systems, models must do more than just produce accurate predictions—they also need to be interpretable, efficient, and practical to deploy. Interpretability is especially important in environmental applications because decision-makers, such as meteorologists and disaster management authorities, must understand why a model makes certain predictions. A 2021 review on interpretable machine learning highlights that models should provide clear explanations of how input features like temperature, humidity, and pressure influence the output. This transparency builds trust and allows experts to validate and refine the system.

In addition to interpretability, forecasting systems require fast iteration and modest computational resources. Unlike large deep learning models that demand high-performance hardware, operational systems often need to run quickly and reliably on limited infrastructure. This is where tree-based models, such as decision trees, random forests, and gradient boosting methods, become highly useful. These models perform well on structured tabular weather data and can capture complex relationships without requiring massive datasets or computing power. They also offer feature importance measures, making them easier to interpret. As a result, such models provide a practical balance between accuracy, efficiency, and explainability in applied weather forecasting systems.

2.5 Positioning of This Project

Glass Weather AI is designed as a practical and educational system rather than a large-scale global forecasting solution. Instead of attempting to compete with complex meteorological infrastructures, it focuses on delivering localized, short-term weather predictions using accessible resources. By relying on public weather APIs, the system gathers real-time and historical data without requiring expensive proprietary datasets. This makes the platform highly suitable for students, researchers, and developers who want to understand how modern weather prediction systems work in a simplified and cost-effective manner. The emphasis is on learning, experimentation, and usability rather than achieving the highest possible forecasting accuracy on a global scale.

A key strength of the system lies in its use of a manageable dataset combined with a gradient boosting model. Gradient boosting is a powerful machine learning technique known for its strong performance on structured, tabular data, which is typical in weather datasets. By training on features such as temperature, humidity, wind speed, and atmospheric pressure, the model can identify patterns and relationships that help predict near-term weather conditions. At the same time, the dataset remains small enough to allow fast training, easy updates, and efficient deployment. This balance ensures that the system remains responsive and adaptable, enabling users to iterate quickly and experiment with different configurations without requiring high computational resources.

The overall design of Glass Weather AI aligns well with modern research and

literature in applied machine learning. It combines data-driven approaches with accessible tools and transparent workflows, making it both practical and interpretable. Unlike black-box systems, the model's predictions can be analyzed and understood, which is especially important in educational settings. The system demonstrates how advanced techniques like machine learning can be integrated into real-world applications without excessive complexity. By prioritizing simplicity, clarity, and efficiency, Glass Weather AI serves as a bridge between theoretical concepts and practical implementation, helping users gain hands-on experience while still producing meaningful and useful weather forecasts.

2.6 Statistical and Baseline Model.

Before the rise of deep learning, weather and time series forecasting primarily depended on classical statistical models. Techniques such as autoregressive (AR) models, moving average (MA) models, and seasonal decomposition were widely used to analyze and predict temporal data. These methods are built on well-defined mathematical assumptions, allowing them to capture important characteristics like trends, seasonality, and temporal dependencies. For example, autoregressive models use past values to predict future observations, while moving averages smooth out noise to reveal underlying patterns. One major advantage of these approaches is their transparency—users can clearly understand how predictions are generated. Additionally, they are computationally efficient, making them suitable for real-time or resource-constrained environments where fast training and quick updates are essential.

Even today, these classical models remain highly relevant, especially as baseline methods in modern forecasting systems. In practical applications, simple models are often used as benchmarks to evaluate the performance of more advanced techniques like machine learning or deep learning. If a sophisticated model fails to outperform a basic seasonal or statistical baseline, it often indicates underlying issues such as data leakage, incorrect feature selection, or flawed evaluation strategies. This makes baseline models an essential diagnostic tool in model development. They help ensure that improvements in accuracy are genuine and not the result of mistakes in data handling or model design. In operational settings, these baselines act as calibration checks, providing a reliable point of comparison for validating more complex systems.

For local and short-term forecasting tasks, simple baseline models can perform surprisingly well. One common example is the persistence model, which assumes that the future value will be the same as the most recent observed value. A slightly more advanced version, seasonal persistence, predicts future values based on observations from the same time in previous cycles, such as the same hour on the previous day. These methods are particularly effective in temperature forecasting because weather variables often exhibit strong autocorrelation over short periods. In the case of Glass Weather AI, the use of lag features allows the gradient boosting model to incorporate similar information implicitly. By learning from past observations, the model captures persistence-like behavior while also integrating additional variables such as humidity or wind speed. This combination enables it to go beyond simple baselines while still benefiting from their core strengths.

2.7 Hybrid and Physics-Informed Approaches.

Modern research in weather and climate forecasting highlights an important limitation of purely data-driven models: they often struggle when the available data is limited, noisy, or does not fully represent real-world conditions. Machine learning models, especially those trained only on historical data, may produce predictions that violate known physical laws—for example, unrealistic temperature fluctuations or inconsistent seasonal patterns. This happens because such models focus on statistical correlations rather than the underlying physical processes governing the atmosphere. To address this issue, researchers have proposed hybrid approaches that combine the strengths of physics-based modeling with the flexibility of machine learning. These methods incorporate physical equations, outputs from numerical weather prediction models, or scientifically derived features into the learning process, ensuring that predictions remain physically consistent while still benefiting from data-driven improvements.

Hybrid models aim to strike a balance between accuracy and reliability. Physics-based models, although computationally expensive, are grounded in well-established scientific principles such as thermodynamics and fluid dynamics. On the other hand, machine learning models are efficient and can capture complex, non-linear relationships in data. By integrating these two approaches, hybrid systems can correct biases in physical models or enhance machine learning predictions with domain knowledge. For example, a hybrid system might use outputs from a physical weather simulation as input features for a machine learning model, allowing it to refine predictions based on both theoretical understanding and observed data. This combination has been widely discussed in recent survey literature as a promising direction for improving forecasting performance, especially in scenarios where data alone is insufficient.

Although Glass Weather AI does not implement a fully developed hybrid framework, it follows a similar philosophy in a simplified and practical way. Instead of directly embedding complex physical equations, the system incorporates physical insights through engineered time-based features such as hour of the day, day of the week, and seasonal indicators. These features capture important patterns like diurnal cycles (daily temperature variations) and seasonal trends, which are fundamentally driven by physical processes such as solar radiation and Earth's rotation. By including these signals, the model gains a basic awareness of real-world behavior without increasing computational complexity. This approach serves as an effective bridge between physical reasoning and machine learning, making Glass Weather AI both interpretable and efficient, while still maintaining reasonable predictive performance for local, short-term forecasting tasks.

CHAPTER 3
FORMULATIONN OF THE PROJECT

3.1 Data Sources and External Services

Modern research in weather and climate forecasting highlights an important limitation of purely data-driven models: they often struggle when the available data is limited, noisy, or does not fully represent real-world conditions. Machine learning models, especially those trained only on historical data, may produce predictions that violate known physical laws-for example, unrealistic temperature fluctuations or inconsistent seasonal patterns. This happens because such models focus on statistical correlations rather than the underlying physical processes governing the atmosphere. To address this issue, researchers have proposed hybrid approaches that combine the strengths of physics-based modeling with the flexibility of machine learning. These methods incorporate physical equations, outputs from numerical weather prediction models, or scientifically derived features into the learning process, ensuring that predictions remain physically consistent while still benefiting from data-driven improvements.

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The system relies on open and well-documented APIs for weather and geocoding. The Open-Meteo forecast API provides current, hourly, and daily forecasts with a flexible set of variables . Open-Meteo also exposes historical weather data through archive endpoints, and provides air quality data through a dedicated endpoint . For optional blended forecasts and air quality, the OpenWeather One Call API 3.0 and the OpenWeather air pollution endpoint are used when an API key is available. To translate coordinates into human-readable locations and to resolve place names into coordinates, the project uses Nominatim search and

reverse geocoding services. explain these things in terms of Machine learning and datasets.

In the context of machine learning, APIs such as Open-Meteo API and OpenWeather One Call API 3.0 act as data sources that continuously supply structured datasets required for model training and prediction. Instead of manually collecting weather records, these APIs provide ready-to-use data including temperature, humidity, wind speed, and precipitation at different time intervals (current, hourly, daily). From a machine learning perspective, this data forms the feature set (input variables) used to train forecasting models. The availability of both real-time and historical data is especially important: historical data (from archive endpoints) is used to train and validate models, while real-time data is used for making live predictions. This setup ensures that the dataset remains dynamic, up-to-date, and scalable without requiring complex data collection pipelines.

Additionally, APIs providing historical weather and air quality data contribute to building rich and diverse datasets. For example, air pollution data from services like OpenWeather Air Pollution API can be integrated as additional features, improving model performance by capturing environmental factors that influence weather patterns. In machine learning terms, this is known as feature enrichment, where adding more relevant variables can help the model learn better patterns and relationships. Similarly, combining multiple APIs (such as Open-Meteo and OpenWeather) can create a blended dataset, improving robustness and reducing dependency on a single data provider. This also helps handle missing data or inconsistencies by cross-referencing multiple sources, which is a common practice in real-world ML systems.

Geocoding services like Nominatim play a crucial role in data preprocessing and feature engineering. Machine learning models typically require numerical inputs, so location names (e.g., cities) must be converted into geographic coordinates (latitude and longitude). Nominatim enables this transformation through forward and reverse geocoding. These coordinates can then be used to fetch location-specific weather data, ensuring that the dataset is spatially accurate. Furthermore, location-based features can be extended to include derived attributes such as elevation, region, or climate zone. Overall, these APIs collectively form a data pipeline that automates data collection, preprocessing, and feature generation—key steps in building an effective and scalable machine learning system for weather forecasting.

3.2 Modeling Strategy.

The model is designed to predict the temperature for the next hour using features created from recent historical data. These features, often called *engineered features*, include past temperature values (lag features), time-related inputs like hour of the day, and possibly other variables such as humidity or wind speed. By using recent observations, the model captures short-term patterns and trends, which are especially important in weather forecasting where conditions change gradually over time.

To perform the prediction, the system uses histogram-based gradient boosting, a type of tree ensemble method. This algorithm builds multiple decision trees

sequentially, where each new tree corrects the errors of the previous ones. It is particularly effective for tabular data and can handle nonlinear relationships and interactions between features without requiring complex preprocessing. Another advantage is that it is computationally efficient and relatively interpretable compared to deep learning models, making it suitable for lightweight applications like Glass Weather AI.
The model predicts one hour ahead and then applies this prediction recursively to estimate further future values. This step-by-step forecasting approach enables short-term predictions while keeping the system simple, fast, and easy to integrate into a user-friendly interface.

3.3 Feature, Engineering & Desing.

The model is specifically designed to predict the temperature for the next hour by using features derived from recent historical data. These engineered features include past temperature values (lag features), time-based inputs such as hour of the day, and possibly other weather variables like humidity or wind speed. By focusing on recent observations, the model effectively captures short-term trends and patterns, which are crucial in weather forecasting where conditions often change gradually and show strong temporal dependency.
To perform the prediction, the system uses histogram-based gradient boosting, an advanced tree ensemble technique. This algorithm builds multiple decision trees in sequence, where each new tree learns from the errors made by the previous ones. It is highly effective for structured or tabular datasets and can naturally capture nonlinear relationships and feature interactions without requiring extensive preprocessing. Additionally, it is computationally efficient and offers a level of interpretability, making it suitable for practical applications where both performance and simplicity are important.
The model generates predictions one step ahead (i.e., the next hour) and then applies this process recursively to produce short-term forecasts over multiple hours. This step-by-step approach keeps the system simple and flexible while still delivering useful predictions. It also makes integration with a user interface easier, allowing users to understand and trust the forecast results without dealing with overly complex modeling techniques.
The App includes meteorological variables and time-based features. Features include:

- Temperature at 2 meters.
- Relative humidity at 2 meters.
- Wind speed at 10 meters.
- Precipitation.
- Hour of day.
- Day of year.
- Month.
- Lagged temperature (1 hour).
- Lagged temperature (24 hours).

3.4 Evaluation Metrics.

Mean Absolute Error (MAE) is used as the primary evaluation metric to measure how well the model predicts temperature. In simple terms, MAE calculates the average of the absolute differences between the predicted values and the actual observed values. This means it looks at how far off each prediction is from the real temperature, ignores whether the error is positive or negative, and then averages these differences. As a result, MAE provides a clear and straightforward indication of the model's accuracy.

One of the main advantages of MAE is its interpretability. Since it is expressed in the same units as the target variable (in this case, degrees of temperature), it is easy to understand. For example, if the MAE is 2°C, it means that, on average, the model's predictions are off by 2 degrees. This makes it very useful for practical applications like weather forecasting, where users need meaningful and understandable performance measures.

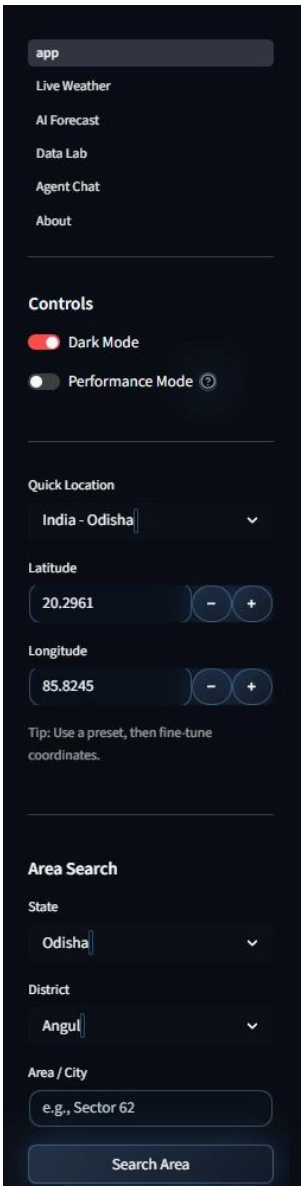
MAE is also widely used in machine learning libraries such as scikit-learn, making it a standard choice for regression problems. It treats all errors equally, without giving extra weight to large errors, which makes it stable and less sensitive to outliers compared to some other metrics. Overall, MAE provides a reliable and easy-to-interpret way to evaluate the performance of temperature prediction models.

3.5 User Interface Design Pattern.

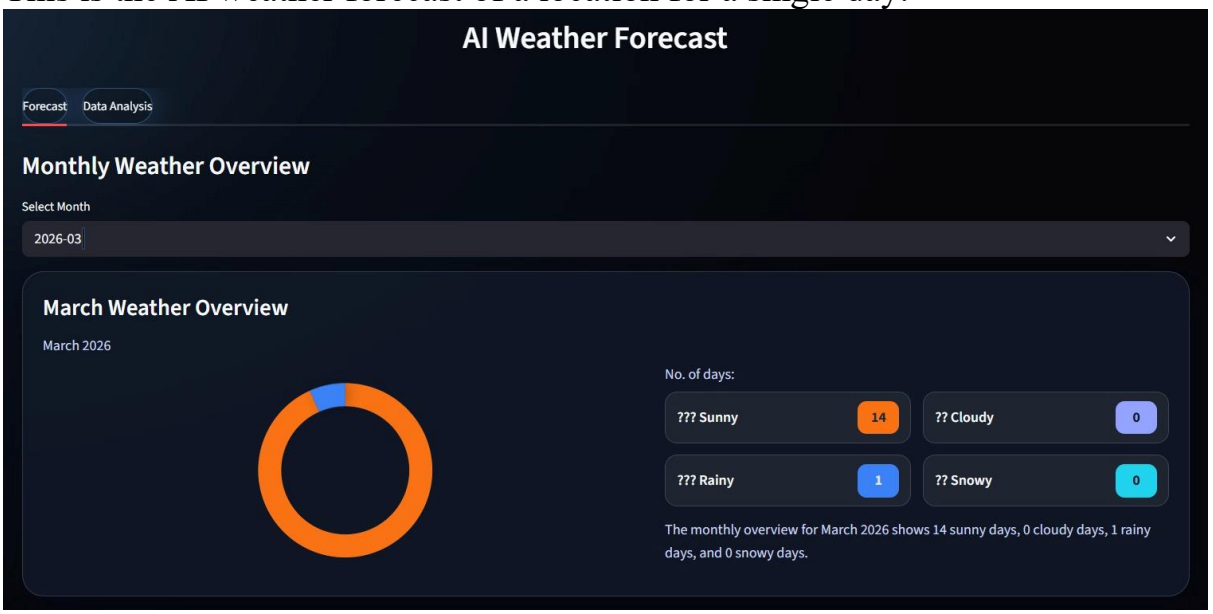
Effective UI design patterns create intuitive, enjoyable user experiences. This section explores patterns relevant to music streaming applications.

This is the home page showing the overall weather forecasting including Temp., Windspeed, Cloud Visibility, Air Quality, Humidity, Perception, Pressure, UV Index.

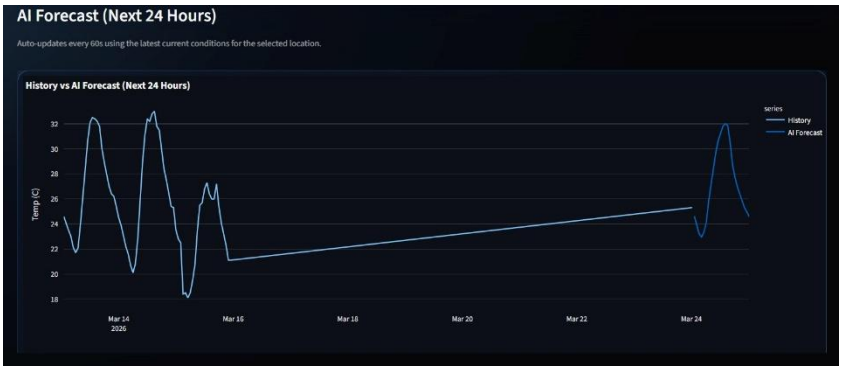
This shows the various menu function provided by the weather prediction AI, like Live weather, AI Forcast, data Lab, Agent Chat, About. It also consist of day and night mode in it.
We can also put the location manually to see the weather condition of that location.



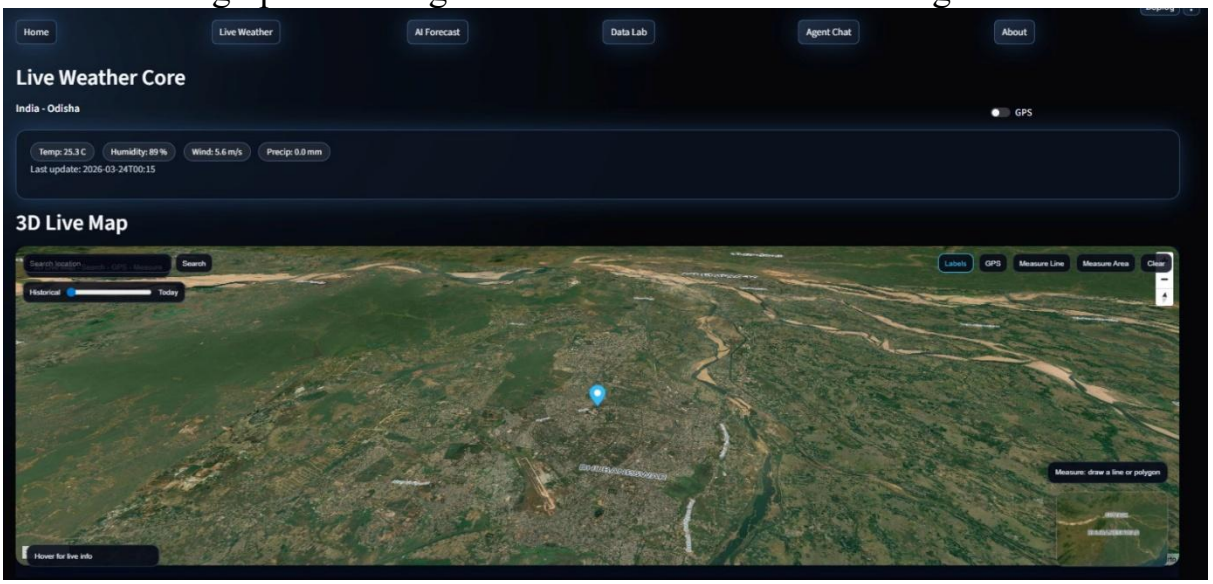
This is the AI weather forecast of a location for a single day.



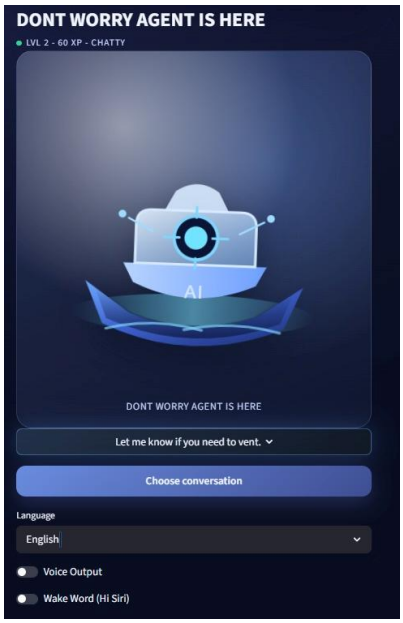
This is the AI Weather forecasting showing the weather condition for a month.

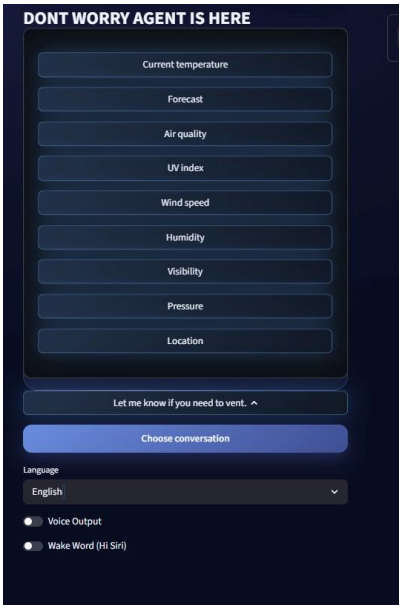


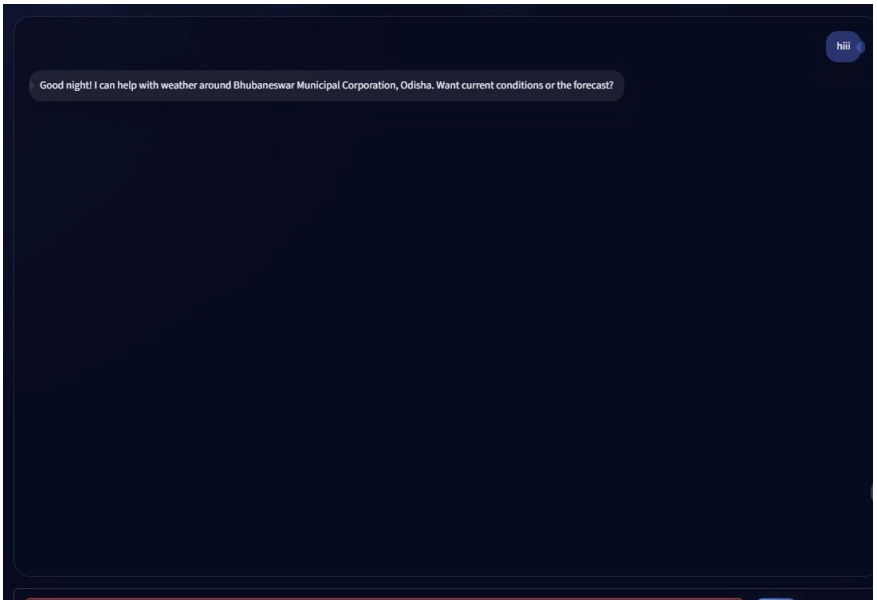
These are the graphs showing the weather conditions according to the datasets.



3D map view options is also available here in which we can see the particular weather condition of a place locating directly in google map. This is the most advance technology given in our project.







An agentic AI mode is given here to help the user to check the weather conditions at a particular place and find all the temperature parameters.

CHAPTER 4
SYSTEM ANALYSIS & DESIGN

4.1 System Requirements

Defining clear system requirements ensures that the Glass Weather AI meets user needs and technical standards. Requirements are categorized into hardware requirements & software requirements.

Hardware Requirements

Development Environment:

- Computer with minimum 4GB RAM
- 10GB free disk space
- Modern processor (Intel i3 or equivalent)
- Internet connection for deployment

User Environment:

- Any device with a modern web browser
- Speakers or headphones for audio output
- Internet connection for streaming
- Minimum 2GB RAM recommended

Software Requirements

Development Tools:

- Jupiter Notebook or similar code editor
- Git for version control
- Modern web browser for testing – SciKit Learn, PyTorch, Tensorflow

Runtime Environment:

- Modern web browser with HTML support
- JavaScript enabled
- Local Storage support
- GPS capability

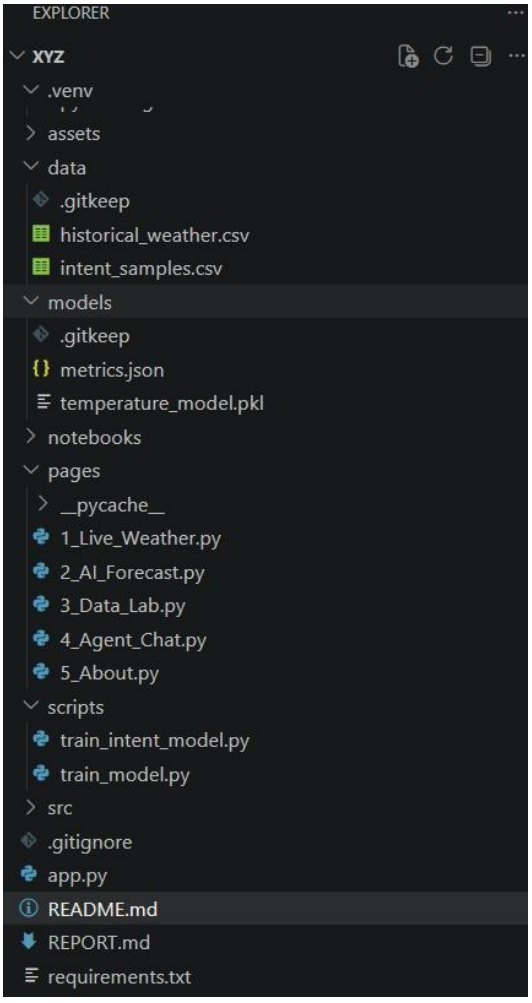
4.2 User Interface design.

The user interface design for the Weather Forecasting AI focuses on replicating Real Weather forecasting's clean, intuitive design while ensuring responsiveness and accessibility.

Design Principles

- **Simplicity**: Clean, uncluttered interface
- **Consistency**: Uniform styling across components
- **Visual Hierarchy**: Clear information structure
- **Feedback**: Immediate response to user actions
- **Accessibility**: Usable by people of all abilities

Project Folder Structure.



CHAPTER 5
CONCLUSION

5.1 Conclusion

This project demonstrates that effective weather forecasting systems can be developed using open data sources and standard Python tools, without requiring large-scale infrastructure or highly complex models. By focusing on local, short-term predictions, the system shows that even compact machine learning solutions can deliver meaningful results. Although it is not intended to replace advanced Numerical Weather Prediction (NWP) systems used by professional meteorological agencies, it serves as a practical and accessible platform for learning and experimentation. This makes it especially valuable for students, developers, and researchers who want to understand how forecasting systems are built and deployed in real-world scenarios.

Another key contribution of the project is its flexibility and potential for extension. The current system focuses mainly on temperature prediction, but it can be easily expanded to include additional weather variables such as humidity, wind speed, or precipitation. Similarly, more advanced machine learning models or hybrid approaches could be integrated to improve performance. The system can also be enhanced with agent-based features, such as automated data updates or intelligent user interactions, making it more interactive and dynamic.

5.2 Summary

Glass Weather AI demonstrates an end-to-end pipeline that integrates public weather services, local data preparation, a lightweight machine learning model, and a responsive user interface. The system is designed for accessibility and reproducibility, enabling both casual users and students to interact with real weather data and experiment with forecasting.

5.3 Future Scope

While the current system meets its core objectives, there are several clear paths for enhancement. First, the model can be extended to predict additional targets such as humidity, wind speed, and precipitation. The existing pipeline already stores these fields in the dataset, so the change would require adding additional supervised targets and training a small set of specialized regressors. Second, the forecasting horizon could be improved by adding probabilistic estimates. For example, the model could output prediction intervals rather than point estimates, improving user understanding of uncertainty. Third, the agentic chat system could be extended with richer natural language understanding and contextual memory. The current rule-based approach provides stability and transparency, but it is not expressive. A hybrid approach that combines intent matching with a small language model could retain deterministic tool routing while improving conversational naturalness.

5.4 Reference

- "Efficacy of Machine Learning in Simulating Precipitation over North India
- IndiaWeatherBench Dataset
- Used advance machine learning tools "SciKit Learn, PyTorch, Tensorflow".

ASSESMENT

SL NO	RUBRICS	FULL MARK	MARKS OBTAINED	REMARKS
1	Understanding the relevance, scope and dimension of the project	10		
2	Methodology	10		
3	Quality of Analysis and Results	10		
4	Interpretations and Conclusions	10		
5	Report	10		
	Total	**50**		

Date: **Signature of the Faculty**